

BEE 4750 Homework 5: Mixed Integer and Stochastic Programming

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Due Date

Thursday, 12/04/24, 9:00pm

Overview

Instructions

- In Problem 1, you will use mixed integer programming to solve a waste load allocation problem.
- In Problem 2, you will formulate a stochastic optimization problem.

Load Environment

The following code loads the environment and makes sure all needed packages are installed. This should be at the start of most Julia scripts.

```
import Pkg
Pkg.activate(@__DIR__)
Pkg.instantiate()
```

```
Activating project at `c:\Users\evant\OneDrive\Documents\GitHub\hw5-evatw`
Warning: The active manifest file has dependencies that were resolved with a different julia
@ nothing C:\Users\evant\OneDrive\Documents\GitHub\hw5-evatw\Manifest.toml:0
Warning: The active manifest file has dependencies that were resolved with a different julia
```

```
@ nothing C:\Users\evant\OneDrive\Documents\GitHub\hw5-ewatw\Manifest.toml:0
Warning: The project dependencies or compat requirements have changed since the manifest wa
It is recommended to `Pkg.resolve()` or consider `Pkg.update()` if necessary.
@ Pkg.API C:\Users\evant\.julia\juliaup\julia-1.12.1+0.x64.w64.mingw32\share\julia\stdlib\v
Warning: The project dependencies or compat requirements have changed since the manifest wa
It is recommended to `Pkg.resolve()` or consider `Pkg.update()` if necessary.
@ Pkg.API C:\Users\evant\.julia\juliaup\julia-1.12.1+0.x64.w64.mingw32\share\julia\stdlib\v
PrecompilingPrecompiling packages... packages...
```

```
MbedTLS
MbedTLS
HTTP
```

```
0 dependencies successfully precompiled in 4 seconds. 249 already precompiled.
```

```
HTTP
```

```
0 dependencies successfully precompiled in 4 seconds. 249 already precompiled.
```

```
using JuMP
using HiGHS
using DataFrames
using GraphRecipes
using Plots
using Measures
using MarkdownTables
```

Problems (Total: 30 Points)

Problem 1 (24 points)

Three cities are developing a coordinated municipal solid waste (MSW) disposal plan. Three disposal alternatives are being considered: a landfill (LF), a materials recycling facility (MRF), and a waste-to-energy facility (WTE). The capacities of these facilities and the fees for operation and disposal are provided below.

- **LF:** Capacity 200 Mg, fixed cost \$2000/day, tipping cost \$50/Mg;
- **MRF:** Capacity 350 Mg, fixed cost \$1500/day, tipping cost \$7/Mg, recycling cost \$40/Mg recycled;
- **WTE:** Capacity 210 Mg, fixed cost \$2500/day, tipping cost \$60/Mg;

The MRF recycling rate is 40%, and the ash fraction of non-recycled waste is 16% and of recycled waste is 14%. Transportation costs are \$1.5/Mg-km, and the relative distances between the cities and facilities are provided in the table below.

City/Facility	Landfill (km)	MRF (km)	WTE (km)
1	5	30	15
2	15	25	10
3	13	45	20
LF	-	32	18
MRF	32	-	15
WTE	18	15	-

The fixed costs associated with the disposal options are incurred only if the particular disposal option is implemented. The three cities produce 100, 90, and 120 Mg/day of solid waste, respectively, with the composition provided in the table below.

Component	% of total mass	Combustion ash (%)	MRF Recycling rate (%)
Food Wastes	15	8	0
Paper & Cardboard	40	7	55
Plastics	5	5	15
Textiles	3	10	10
Rubber, Leather	2	15	0
Wood	5	2	30
Yard Wastes	18	2	40
Glass	4	100	60
Ferrous	2	100	75
Aluminum	2	100	80
Other Metal	1	100	50
Miscellaneous	3	70	0

The information in the above table will help you determine the overall recycling and ash fractions. Note that the recycling residuals, which may be sent to either landfill or the WTE, have different ash content than the ash content of the original MSW. You will need to determine these fractions to construct your mass balance constraints.

Reminder: Use `round(x; digits=n)` to report values to the appropriate precision!

Problem 1.1

Based on the information above, calculate the overall recycling and ash fractions for the waste produced by each city.

Recycling fraction would be the percentage of total mass recycled with MRF, ash fraction would be the remaining ash from the non-recycled component w.r.t the total mass.

Component	% of total mass	Combustion ash (%)	MRF Recy- cling rate (%)	Non- recycled amount (%)	Recycle fraction (%)	Ash fraction (%)
Food Wastes	15	8	0	15	0	1.20
Paper & Cardboard	40	7	55	18	22	1.26
Plastics	5	5	15	4.25	0.75	0.21
Textiles	3	10	10	2.70	0.30	0.27
Rubber, Leather	2	15	0	2	0	0.3
Wood	5	2	30	3.5	1.5	0.07
Yard Wastes	18	2	40	10.8	7.2	0.216
Glass	4	100	60	1.6	2.4	1.6
Ferrous	2	100	75	0.5	1.5	0.5
Aluminum	2	100	80	0.4	1.6	0.4
Other Metal	1	100	50	0.5	0.5	0.5
Miscellaneous	3	70	0	3	0	2.1

The total recycle fraction from each city would be 37.75, while the total ash fraction would be 8.62.

Problem 1.2

What are the decision variables for your optimization problem? Provide notation and variable meaning.

The decision variables for this optimization problem would be the mass of waste going into each facility f from each city i in Mg. This problem will assume that the facilities are open 24/7. There is also the mass of recycle R from each city to be processed by the MRF.

$$x_{i,f}, R_i$$

Problem 1.3

Formulate the objective function. Make sure to include any needed derivations or justifications for your equation(s).

The objective of this function is to minimize the total daily cost of the waste disposal system. The total daily cost would be composed of the fixed costs for operating the facilities in dollars per day, tipping costs in dollars per Mg, transportation costs in dollars per Mg-km, and recycling costs in dollars per Mg recycled.

$$Total = Fixed + Tipping + Transportation + Recycling$$

The fixed cost F would be $2000 + 1500 + 2500$ dollars per day. The tipping cost T would be the cost multiplied by the mass of waste entering each facility. The transportation cost Tr would be the distance from each city to facility multiplied by transportation cost multiplied by mass of waste being transported. The recycling cost Re would be the recycling processing cost multiplied by mass recycled R from each city.

$$\begin{aligned}
 F &= 2000 + 1500 + 2500 \\
 T &= \sum_{i,f} t_f \times x_{i,f} \\
 Tr &= \sum_{i,f} d_{i,f} \times 1.5x_{i,f} \\
 Re &= \sum_i R_i \times 40 \\
 Total &= 6000 + \sum_{i,f} t_f \times x_{i,f} + \sum_{i,f} d_{i,f} \times 1.5x_{i,f} + \sum_i R_i \times 40
 \end{aligned}$$

Problem 1.4

Derive all relevant constraints. Make sure to include any needed justifications or derivations.

The relevant constraints would be that disposal cannot exceed the maximum capacity of each facility due to system constraints. So LF cannot exceed 200 Mg, MRF cannot exceed 350 Mg, WTE cannot exceed 210 Mg. Additionally, the waste consumed and residuals produced from each city must match the amount disposed daily (e.g. 100, 90 and 120 for cities 1-3). Any of the relevant decision variables must be non-negative for conservation of mass.

Problem 1.5

Find the optimal solution (using JuMP to solve the problem). Report the optimal objective value.

```
# Build and solve the MSW allocation MILP
model = Model(HiGHS.Optimizer)

# Data
cities = [1, 2, 3]
supply = [100.0, 90.0, 120.0] # Mg/day
facilities = ["LF", "MRF", "WTE"]
```

```

num_facilities = 3

cap = Dict("LF" => 200.0, "MRF" => 350.0, "WTE" => 210.0)
fixed = Dict("LF" => 2000.0, "MRF" => 1500.0, "WTE" => 2500.0)
tipping = Dict("LF" => 50.0, "MRF" => 7.0, "WTE" => 60.0)
recycle_rate = 0.40
recycle_proc_cost = 40.0
transport_cost_per_Mg_km = 1.5

# City to facility distances (km) - rows: cities 1,2,3; cols: LF, MRF, WTE
dist_cf = [5.0 30.0 15.0;
            15.0 25.0 10.0;
            13.0 45.0 20.0]

# Facility to facility distances (km) - for residual transport from MRF
dist_ff = Dict(("MRF", "LF") => 32.0, ("MRF", "WTE") => 15.0)

# Decision Variables
# x[i,j] = mass (Mg) from city i to facility j (1=LF, 2=MRF, 3=WTE)
@variable(model, x[i in cities, j in 1:num_facilities] >= 0)

# r[i,j] = residual mass from city i's MRF processing sent to facility j (1=LF, 3=WTE)
# r[i,1] = to LF, r[i,3] = to WTE
@variable(model, r[i in cities, j in [1, 3]] >= 0)

# y[j] = binary: 1 if facility j is open (1=LF, 2=MRF, 3=WTE)
@variable(model, y[j in 1:num_facilities], Bin)

# Constraints

# Supply balance: each city's waste must go somewhere (to LF, MRF, or WTE)
for i in cities
    @constraint(model, sum(x[i, j] for j in 1:num_facilities) == supply[i])
end

# MRF residual balance: non-recycled mass from MRF must go to LF or WTE
for i in cities
    @constraint(model, r[i, 1] + r[i, 3] == (1.0 - recycle_rate) * x[i, 2])
end

# Facility capacity constraints (with facility open indicator)
@constraint(model, sum(x[i, 1] for i in cities) + sum(r[i, 1] for i in cities) <= cap["LF"] :

```

```

@constraint(model, sum(x[i, 2] for i in cities) <= cap["MRF"] * y[2]) # MRF
@constraint(model, sum(x[i, 3] for i in cities) + sum(r[i, 3] for i in cities) <= cap["WTE"])

# Objective: minimize total daily cost
# Fixed costs (incurred only if facility is open)
fixed_cost = 2000 * y[1] + 1500 * y[2] + 2500 * y[3]

# Tipping costs (applied to mass entering each facility)
tipping_cost = 50 * (sum(x[i, 1] for i in cities) + sum(r[i, 1] for i in cities)) + # LF
               7 * sum(x[i, 2] for i in cities) + # MRF
               60 * (sum(x[i, 3] for i in cities) + sum(r[i, 3] for i in cities)) # WTE

# Transportation costs: city to facility
transport_city_fac = sum(1.5 * dist_cf[i, j] * x[i, j] for i in cities, j in 1:num_facilities)

# Transportation costs: residual from MRF to LF or WTE
transport_residual = 1.5 * dist_ff[("MRF", "LF")] * sum(r[i, 1] for i in cities) +
                    1.5 * dist_ff[("MRF", "WTE")] * sum(r[i, 3] for i in cities)

# Recycling processing cost (applies to recycled mass from MRF)
recycle_cost = 40 * 0.40 * sum(x[i, 2] for i in cities)

@objective(model, Min, fixed_cost + tipping_cost + transport_city_fac + transport_residual +

# Solve
optimize!(model)

```

Running HiGHS 1.12.0 (git hash: 755a8e027): Copyright (c) 2025 HiGHS under MIT licence terms
MIP has 9 rows; 18 cols; 36 nonzeros; 3 integer variables (3 binary)

Coefficient ranges:

Matrix	[6e-01, 4e+02]
Cost	[6e+01, 2e+03]
Bound	[1e+00, 1e+00]
RHS	[9e+01, 1e+02]

Presolving model

9 rows, 18 cols, 36 nonzeros 0s

6 rows, 15 cols, 33 nonzeros 0s

6 rows, 15 cols, 33 nonzeros 0s

Presolve reductions: rows 6(-3); columns 15(-3); nonzeros 33(-3)

Solving MIP model with:

6 rows

15 cols (3 binary, 0 integer, 0 implied int., 12 continuous, 0 domain fixed)
33 nonzeros

Src: B => Branching; C => Central rounding; F => Feasibility pump; H => Heuristic;
I => Shifting; J => Feasibility jump; L => Sub-MIP; P => Empty MIP; R => Randomized round
S => Solve LP; T => Evaluate node; U => Unbounded; X => User solution; Y => HiGHS solution
Z => ZI Round; l => Trivial lower; p => Trivial point; u => Trivial upper; z => Trivial

Src	Nodes		B&B Tree		Objective Bounds			Dynamic Cuts	
	Proc.	InQueue	Leaves	Expl.	BestBound	BestSol	Gap	Cuts	
J	0	0	0	0.00%	-inf	43346	Large	0	
S	0	0	0	0.00%	14030	25750	45.51%	0	

66.7% inactive integer columns, restarting

Model after restart has 6 rows, 12 cols (0 bin., 0 int., 0 impl., 12 cont., 0 dom.fix.), and

0	0	0	0.00%	24559.52381	25750	4.62%	0
1	0	1	100.00%	25750	25750	0.00%	0

Solving report

Status Optimal
Primal bound 25750
Dual bound 25750
Gap 0% (tolerance: 0.01%)
P-D integral 0.000300166406296
Solution status feasible
25750 (objective)
0 (bound viol.)
0 (int. viol.)
0 (row viol.)
Timing 0.00
Max sub-MIP depth 0
Nodes 1
Repair LPs 0
LP iterations 8
0 (strong br.)
0 (separation)
0 (heuristics)

Results
using Printf

```

obj_value = objective_value(model)
@printf("Optimal Total Daily Cost: %.2f USD\n\n", obj_value)

# Facility usage
@printf("LF open:  %d (capacity utilized: %.2f / 200.0 Mg)\n", Int(value(y[1])),
        value(sum(x[i, 1] for i in cities) + sum(r[i, 1] for i in cities)))
@printf("MRF open: %d (capacity utilized: %.2f / 350.0 Mg)\n", Int(value(y[2])),
        value(sum(x[i, 2] for i in cities)))
@printf("WTE open: %d (capacity utilized: %.2f / 210.0 Mg)\n\n", Int(value(y[3])),
        value(sum(x[i, 3] for i in cities) + sum(r[i, 3] for i in cities)))

# Flows from cities
println("Detailed Flows (Mg/day):")
@printf("%-8s | %-10s | %-10s | %-10s\n", "City", "→ LF", "→ MRF", "→ WTE")
println(repeat("-", 45))
for i in cities
    @printf("City %d    | %10.2f | %10.2f | %10.2f\n", i, value(x[i, 1]), value(x[i, 2]), value(x[i, 3]))
end

println("\nMRF Residual Flows (Mg/day):")
@printf("%-8s | %-15s | %-15s\n", "City", "Residual → LF", "Residual → WTE")
println(repeat("-", 40))
for i in cities
    @printf("City %d    | %15.2f | %15.2f\n", i, value(r[i, 1]), value(r[i, 3]))
end

# Summary
println("\nTotal Facility Flows (Mg/day):")
total_lf = value(sum(x[i, 1] for i in cities) + sum(r[i, 1] for i in cities))
total_mrf = value(sum(x[i, 2] for i in cities))
total_wte = value(sum(x[i, 3] for i in cities) + sum(r[i, 3] for i in cities))
@printf("LF:  %.2f\n", total_lf)
@printf("MRF: %.2f\n", total_mrf)
@printf("WTE: %.2f\n", total_wte)

# Cost breakdown
println("\nCost Breakdown (USD):")
fixed_cost_val = value(fixed_cost);
tipping_cost_val = value(tipping_cost);
transport_city_val = value(transport_city_fac);
transport_resid_val = value(transport_residual);
recycle_cost_val = value(recycle_cost);

```

```

@printf("Fixed costs:                %.2f\n", fixed_cost_val)
@printf("Tipping costs:              %.2f\n", tipping_cost_val)
@printf("City→Facility transport:    %.2f\n", transport_city_val)
@printf("Residual transport:          %.2f\n", transport_resid_val)
@printf("Recycling processing:         %.2f\n", recycle_cost_val)

```

Optimal Total Daily Cost: 25750.00 USD

LF open: 1 (capacity utilized: 200.00 / 200.0 Mg)
MRF open: 0 (capacity utilized: 0.00 / 350.0 Mg)
WTE open: 1 (capacity utilized: 110.00 / 210.0 Mg)

Detailed Flows (Mg/day):

City	→ LF	→ MRF	→ WTE
City 1	100.00	0.00	0.00
City 2	0.00	0.00	90.00
City 3	100.00	-0.00	20.00

MRF Residual Flows (Mg/day):

City	Residual → LF	Residual → WTE
City 1	0.00	0.00
City 2	0.00	0.00
City 3	0.00	0.00

Total Facility Flows (Mg/day):

LF: 200.00
MRF: 0.00
WTE: 110.00

Cost Breakdown (USD):

Fixed costs: 4500.00
Tipping costs: 16600.00
City→Facility transport: 4650.00
Residual transport: 0.00
Recycling processing: 0.00
LF open: 1 (capacity utilized: 200.00 / 200.0 Mg)
MRF open: 0 (capacity utilized: 0.00 / 350.0 Mg)
WTE open: 1 (capacity utilized: 110.00 / 210.0 Mg)

Detailed Flows (Mg/day):

City	→ LF	→ MRF	→ WTE

City 1	100.00	0.00	0.00
City 2	0.00	0.00	90.00
City 3	100.00	-0.00	20.00

MRF Residual Flows (Mg/day):

City	Residual → LF	Residual → WTE

City 1	0.00	0.00
City 2	0.00	0.00
City 3	0.00	0.00

Total Facility Flows (Mg/day):

LF: 200.00

MRF: 0.00

WTE: 110.00

Cost Breakdown (USD):

Fixed costs:	4500.00
Tipping costs:	16600.00
City→Facility transport:	4650.00
Residual transport:	0.00
Recycling processing:	0.00

Problem 1.6

Draw a diagram showing the flows of waste between the cities and the facilities. Which facilities (if any) will not be used? Does this solution make sense?

From the waste flow diagram, only the MRF facility is not used. Although this initially appears strange, this is because the MRF facility only recycles a fraction of the incoming waste. Non recyclable waste would still have to go through the LF or WTE facilities, adding extra transportation costs than just sending waste to those facilities in the first place. Sending waste to the MRF facility would only be prioritized if the recycling rate was higher or there are lower transportation costs to justify the recycling process.

Problem 2 (6 points)

Consider a two-period economic dispatch problem, based on the multi-period example from Lecture 14 (on 10/29). The generator data, including ramping constraints for each generator,

is provided in ‘data/generators.csv.’ In period 1, the demand is $d_1 = 1100\text{MW}$. In period 2, the demand is projected to be $d_2 = 1200\text{MW}$, but there is a 25% probability that it is 1500 . In the first period, the solar capacity factor is 0.9 and the wind capacity factor is 0.45, but in the second period, there is some uncertainty: the forecasted solar and wind capacity factors are 0.95 and 0.4, respectively, but there is a 30% probability that they are 0.75 and 0.5. Your goal is to identify how to dispatch your generators to minimize the cost of meeting demand.

Problem 2.1

Draw a scenario tree for this problem.

For Period 1, demand is at 1100 MW with solar capacity factor of 0.9 and wind capacity factor of 0.45. These are guaranteed to happen.

For Period 2, there is a 75% probability that demand is at 1200 MW, and a 25% probability that demand is 1500 MW. Additionally, there is a 70% probability that solar and wind capacity factors are 0.95 and 0.4, and a 30% probability that those factors are 0.75 and 0.5.

Problem 2.2

Formulate a stochastic linear program for this problem based on your scenario tree from Problem 2.1 and the data in `data/generators.csv`.

Known: - $d_1 = 1100$ MW (period 1, deterministic) - $d_2^A = 1200$ MW (scenario A) - $d_2^B = 1500$ MW (scenario B)

- Scenario A: $c_s^A = 0.95$ (solar), $c_w^A = 0.4$ (wind)
- Scenario B: $c_s^B = 0.75$ (solar), $c_w^B = 0.5$ (wind)
- Period 1 (deterministic): $c_s^1 = 0.9$, $c_w^1 = 0.45$

Generator data (from `generators.csv`): - $P_g^{\min}$ = minimum power output (MW) - $P_g^{\max}$ = maximum power capacity (MW) - c_g = variable cost (\$/MWh) - R_g = ramp rate (MW/hour, assuming 1-hour periods)

Renewable capacities: - Wind: 300 MW - Solar: 500 MW

Decision Variables

Stage 1 (Period 1 - Deterministic): - $p_{g,1}$ = power output from generator g in period 1 (MW)

Stage 2 (Period 2 - Scenario-dependent): - $p_{g,2}^s$ = power output from generator g in period 2 under scenario $s \in \{A, B\}$ (MW) - $w_{g,2}^s$ = renewable generation (wind/solar) in period 2 under scenario s (MW)

The objective function would be minimizing the total operational costs while ensuring that capacity is met.

Minimize expected total operational cost:

$$\min \sum_{g \in G} c_g \cdot p_{g,1} + \sum_{s \in S} \pi_s \left(\sum_{g \in G} c_g \cdot p_{g,2}^s \right)$$

where: - $\pi_A = 0.70$ (probability of scenario A) - $\pi_B = 0.30$ (probability of scenario B) - c_g = variable cost of generator g (\$/MWh) - Renewable generators (Wind, Solar) have $c_g = 0$ (no fuel cost)

Constraints

Period 1 Constraints (Deterministic):

1. Demand balance:

$$\sum_{g \in G} p_{g,1} = d_1 = 1100 \text{ MW}$$

2. Generator capacity limits:

$$P_g^{\min} \leq p_{g,1} \leq P_g^{\max} \quad \forall g \in G$$

3. Renewable generation limits (period 1):

$$p_{\text{Wind},1} \leq c_w^1 \cdot 300 = 0.45 \cdot 300 = 135 \text{ MW}$$

$$p_{\text{Solar},1} \leq c_s^1 \cdot 500 = 0.9 \cdot 500 = 450 \text{ MW}$$

Period 2 Constraints (Scenario-dependent for $s \in \{A, B\}$):

4. Demand balance (scenario-dependent):

$$\sum_{g \in G} p_{g,2}^s = d_2^s \quad \forall s \in S$$

where $d_2^A = 1200$ MW and $d_2^B = 1500$ MW

5. Generator capacity limits (scenario-dependent):

$$P_g^{\min} \leq p_{g,2}^s \leq P_g^{\max} \quad \forall g \in G, s \in S$$

6. Renewable generation limits (scenario-dependent):

$$p_{\text{Wind},2}^A \leq c_w^A \cdot 300 = 0.4 \cdot 300 = 120 \text{ MW}$$

$$p_{\text{Wind},2}^B \leq c_w^B \cdot 300 = 0.5 \cdot 300 = 150 \text{ MW}$$

$$p_{\text{Solar},2}^A \leq c_s^A \cdot 500 = 0.95 \cdot 500 = 475 \text{ MW}$$

$$p_{\text{Solar},2}^B \leq c_s^B \cdot 500 = 0.75 \cdot 500 = 375 \text{ MW}$$

Ramping Constraints (between periods):

7. Ramp limits between period 1 and period 2 (scenario-dependent):

$$-R_g \leq p_{g,2}^s - p_{g,1} \leq R_g \quad \forall g \in G, s \in S$$

where R_g is the ramp rate (MW/hour) for generator g

Example constraints for specific generators: - NG CCGT: $-100 \leq p_{\text{NG CCGT},2}^s - p_{\text{NG CCGT},1} \leq 100$ - NG CT: $-200 \leq p_{\text{NG CT},2}^s - p_{\text{NG CT},1} \leq 200$ - Wind/Solar: ± 300 MW

Non-negativity:

$$p_{g,t}^s \geq 0 \quad \forall g \in G, t \in T, s \in S$$

Problem Characteristics

- **Type:** Two-stage stochastic linear program
- **Number of scenarios:** 2 (A and B)
- **Decision stages:** 2 (Period 1 is first-stage; Period 2 is second-stage)
- **Variables:**
 - Period 1: 7 generators
 - Period 2: 7 generators $\times$ 2 scenarios = 14 variables
 - **Total: 21 continuous variables**
- **Constraints:**
 - Period 1: 1 demand + 7 capacity + 2 renewable = 10 constraints
 - Period 2 (each scenario): 1 demand + 7 capacity + 2 renewable + 7 ramp = 17 constraints per scenario
 - **Total: $10 + 2(17) = 44$ constraints**

Key Insights

1. **First-stage commitment** (Period 1) must be feasible for ALL scenarios
2. **Ramping constraints** link Periods 1 and 2, restricting how much generators can change output
3. **Recourse actions** (Period 2) adapt to realized renewable capacity factors and demand
4. **Expected cost minimization** weights scenarios by their probabilities

References

List any external references consulted, including classmates.